

Physics Assignment 1

Q1) $\Delta x_1 = 40 \text{ km}$

$$\Delta t_1 = ?$$

$$\Delta v_1 = 30 \text{ km/hr}$$

$$\text{so } t_1 = \frac{\Delta x_1}{\Delta v_1} = \frac{40}{30} = 1.33 \text{ hr}$$

$$\Delta x_2 = 40 \text{ km}$$

$$\Delta t_2 = ?$$

$$\Delta v_2 = 60 \text{ km/hr}$$

$$\text{so } t_2 = \frac{\Delta x_2}{\Delta v_2} = \frac{40}{60} = 0.67 \text{ hr}$$

$$\text{avg velocity} = \frac{\Delta x_1 + \Delta x_2}{\Delta t_1 + \Delta t_2}$$

$$= \frac{40 + 40}{1.33 + 0.67}$$

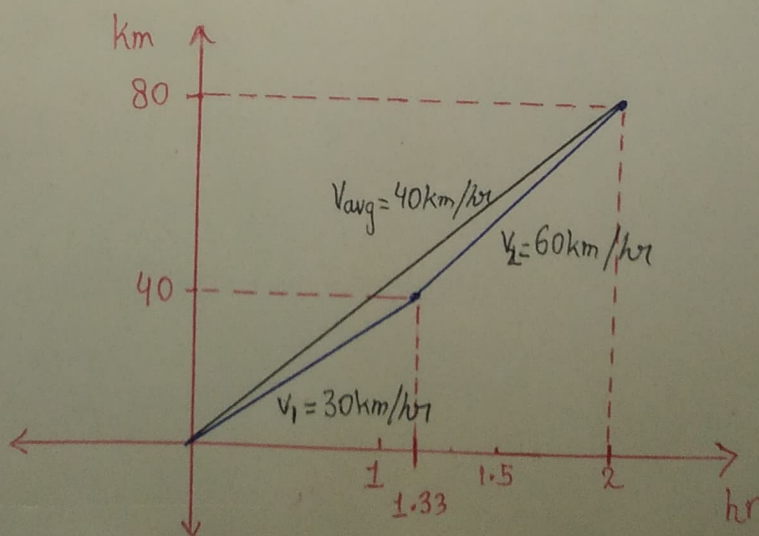
$$= 40 \text{ km/hr} \quad \text{Ans}$$

b) Average speed = 40 km/hr

The avg speed is equal to the magnitude of average velocity because the automobile is moving in a straight line and the same direction

Then the distance traveled becomes equal to the magnitude of displacement vector.

c)



Q2) $U = 30 \text{ m/s}$

$V \text{ at max height} = 0 \text{ m/s}$

$g = 9.8 \text{ m/s}^2$

- Time taken to reach max height

$$\Rightarrow V = U + gt$$

$$0 = 30 + (-9.8)t$$

$$\frac{-30}{-9.8} = t$$

$$3.06 \text{ s} = t$$

Ans

- Total time in the air

$$t_T = 2(3.06)$$

$$t_T = 6.12 \text{ s}$$

Ans

Q3) $X = 3t - 4t^2 + t^3$

a) $t = 1 \text{ s}$

$$X = 3(1) - 4(1)^2 + (1)^3$$

$$X = 0 \text{ m}$$

d) $t = 4 \text{ s}$

$$X = 3(4) - 4(4)^2 + (4)^3$$

$$X = 12 \text{ m}$$

b) $t = 2 \text{ s}$

$$X = 3(2) - 4(2)^2 + (2)^3$$

$$X = -2 \text{ m}$$

~~e) $t = 4 \text{ s}$~~

e) Displacement = $x(4) - x(0)$

$$= 12 \text{ m} - 0 \text{ m}$$

$$= 12 \text{ m}$$

c) $t = 3 \text{ s}$

$$X = 3(3) - 4(3)^2 + (3)^3$$

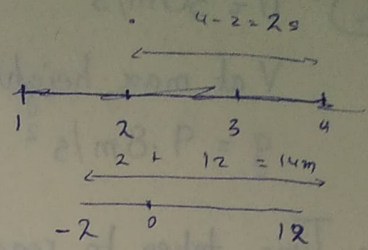
$$X = 0 \text{ m}$$

f) $\text{avg velocity} = \frac{\text{Total Displacement}}{\text{Total Time}}$

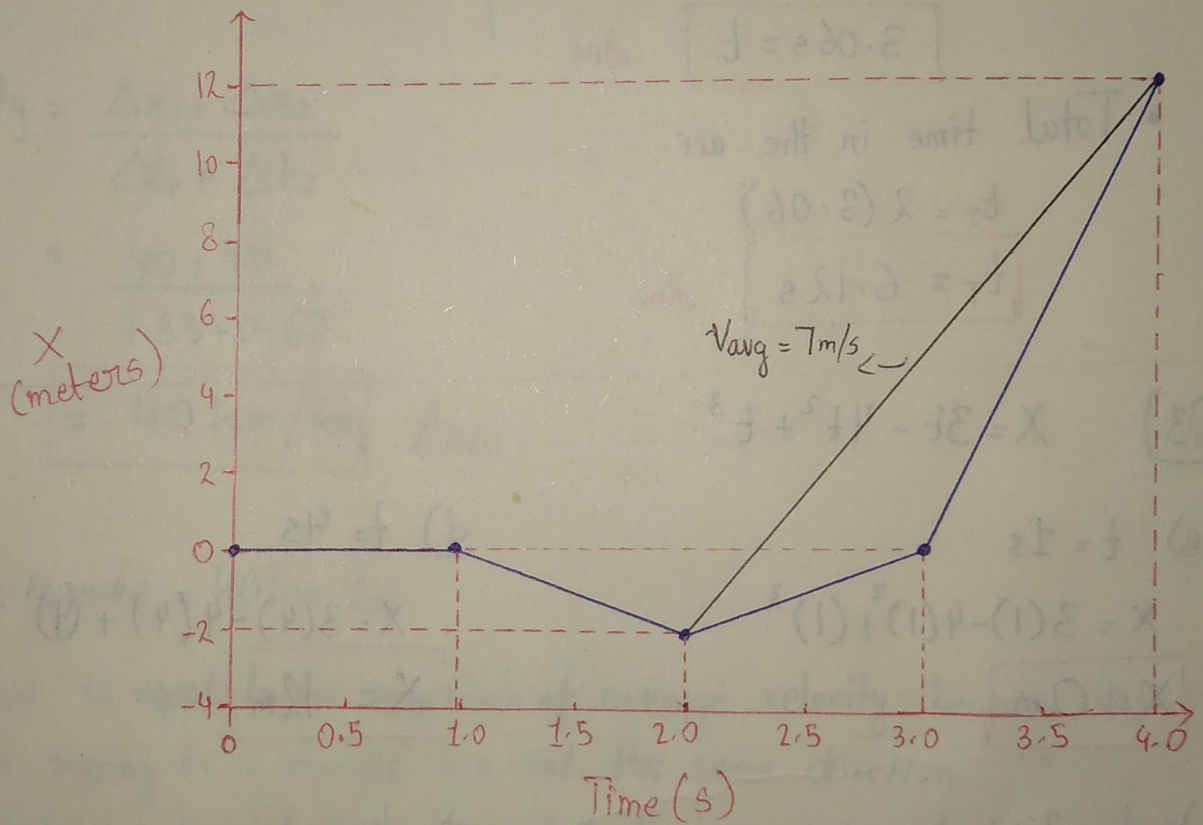
$$V_{\text{avg}} = \frac{12 - (-2)}{4 - 2}$$

$$V_{\text{avg}} = \frac{14}{2}$$

$$V_{\text{avg}} = 7 \text{ m/s}$$



g)



Q4) component of particle B

$$B_y = a \cos \theta = 0.4 \cos \theta$$

$$B_x = a \sin \theta = 0.4 \sin \theta$$

motion of B in y direction

$$S = ut + \frac{1}{2} at^2$$

$$30 = 0 + \frac{1}{2} (0.4 \cos \theta) t^2$$

$$30 = 0.2 \cos \theta t^2 \rightarrow \text{eq 1}$$

motion of B in x direction

$$S = ut + \frac{1}{2} at^2$$

$$x_B = 0 + \frac{1}{2} (0.4 \sin \theta) t^2$$

$$x_B = 0.2 \sin \theta t^2$$

for collision to occur

$$x_A = x_B$$

$$3.0t = 0.2 \sin \theta t^2$$

$$3 = 0.2 \sin \theta t$$

$$t = \frac{3}{0.2 \sin \theta}$$

$$t = \frac{15}{\sin \theta}$$

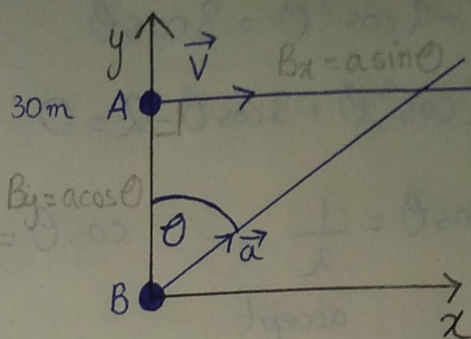
substitute t in eq 1

$$30 = 0.2 \cos \theta \left(\frac{15}{\sin \theta} \right)^2$$

$$30 = \frac{45 \cos \theta}{\sin^2 \theta}$$

$$30 \sin^2 \theta = 45 \cos \theta$$

$$\frac{2}{3} (1 - \cos^2 \theta) = \cos \theta$$



motion of A in x-direction

$$V_A = \frac{x_A}{t}$$

$$x_A = V_A t$$

$$x_A = 3.0t$$

$$2 - 2\cos^2\theta = 3\cos\theta$$

$$2\cos^2\theta + 3\cos\theta - 2 = 0$$

$$\cos\theta = \frac{1}{2} \quad \cos\theta = -2 \text{ reject}$$

accept

for θ

$$\theta = \cos^{-1}\left(\frac{1}{2}\right)$$

$$\theta = 60^\circ \quad \text{Ans}$$

Q5) $\vec{r} = (2.00t^3 - 5.00t)\mathbf{i} + (6.00 - 7.00t^4)\mathbf{j}$

a) vector $\mathbf{r}$ for $t = 2\text{s}$

$$\vec{r} = (2(2)^3 - 5(2))\mathbf{i} + (6 - 7(2)^4)\mathbf{j}$$

$$\vec{r} = (16 - 10)\mathbf{i} + (6 - 112)\mathbf{j}$$

$$\vec{r} = 6\mathbf{i} - 106\mathbf{j} \quad \text{meter} \quad \text{Ans}$$

b) vector $\mathbf{v}$ for $t = 2\text{s}$

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (6t^2 - 5)\mathbf{i} + (-28t^3)\mathbf{j}$$

$$= (6t^2 - 5)\mathbf{i} + (-28t^3)\mathbf{j}$$

$$= (6t^2 - 5)\mathbf{i} - (28t^3)\mathbf{j}$$

substitute $t = 2\text{s}$

$$\mathbf{v}(2) = (6(2)^2 - 5)\mathbf{i} - (28(2)^3)\mathbf{j}$$

$$\vec{v} = 19\mathbf{i} - 224\mathbf{j} \quad \text{m/s} \quad \text{Ans}$$

c) vector a for $t = 2s$

$$a(t) = \frac{dv}{dt} ((6t^2 - 5)\hat{i} - 28t^3\hat{j})$$
$$= (12t)\hat{i} - 84t^2\hat{j}$$

substitute $t = 2s$

$$\vec{a} = (12(2))\hat{i} - 84(2^2)\hat{j}$$

$$\vec{a} = 24\hat{i} - 336\hat{j} \text{ m/s}^2$$

d) for angle b/w positive direction of the x -axis and a line tangent to the particle.

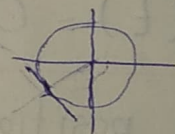
$$v(t) = (6t^2 - 5)\hat{i} - (28t^3)\hat{j}$$

$$v(2) = 19\hat{i} - 224\hat{j}$$

$$\text{angle} \Rightarrow \tan \theta = \left[\frac{-224}{19} \right] =$$

$$\theta = \tan^{-1} \left[\frac{-224}{19} \right]$$

$$\theta = -85.16^\circ$$



Q6)

for t (through vertical component)

$$s = \frac{1}{2} at^2$$

$$\sqrt{\frac{2 \times 2}{9.81}} = \sqrt{t^2}$$

$$0.6385 = t$$

$$0.64 s = t$$

for v

$$v = \frac{x}{t}$$

$$v = \frac{10}{0.64}$$

$$v = 15.62 \text{ m/s}$$

for centripetal acceleration

$$a_c = \frac{v^2}{r}$$

$$a_c = \frac{15.62^2}{1.5} = 162.7 \text{ m/s}^2$$

~~for v~~

~~$$v = \frac{x}{t} = \frac{10}{0.64} = 15.62 \text{ m/s}$$~~

~~for a~~

~~$$a_c = \frac{v^2}{r} = \frac{15.62^2}{1.5} =$$~~

Q7a) for velocity as function of time

$$v(t) = \int a(t) dt$$

$$v(t) = \int 1.5t - 0.12t^2 dt$$

$$v(t) = \frac{1.5t^2}{2} - \frac{0.12t^3}{3} + C$$

for C

motorcycle at rest so $v=0$ at $t=0$

$$v(0) = 0$$

$$0 = \frac{1.5(0)^2}{2} - \frac{0.12(0)^3}{3} + C$$

$$0 = C$$

$$\therefore v(t) = 0.75t^2 - 0.04t^3$$

for position as function of time

$$x(t) = \int v(t) dt$$

$$x(t) = \int 0.75t^2 - 0.04t^3 dt$$

$$x(t) = \frac{0.75t^3}{3} - \frac{0.04t^4}{4} + C$$

as motorcycle starts at origin so $x=0$ at $t=0$

$$x(0) = 0$$

$$0 = \frac{0.75(0)^3}{3} - \frac{0.04(0)^4}{4} + C$$

$$0 = C$$

$$\therefore x(t) = 0.25t^3 - 0.01t^4$$

b) velocity is max when acceleration is 0

$$a(t) = 0$$

$$1.5t - 0.12t^2 = 0$$

$$t(1.5 - 0.12t) = 0$$

$$t = 0$$

(reject)
as it's starting
point

$$-0.12t + 1.5 = 0$$

$$t = \frac{1.5}{0.12}$$

$$t = 12.5s$$

for velocity at $t = 12.5s$

$$v(12.5) = 0.75(12.5)^2 - 0.04(12.5)^3$$

$$v = 39.06 \text{ m/s}$$

Q8a

$$v = u + at$$

$$v = 6 + 9.81(2)$$

$$v = 25.62 \text{ m/s}$$

$$b) s = ut + \frac{1}{2}at^2$$

$$s = 6(2) + \frac{1}{2}(9.81)(2)^2$$

$$s = 31.62 \text{ m}$$

$$c) v^2 = u^2 + 2as$$

$$\sqrt{v^2} = \sqrt{6^2 + 2(9.81)(10)}$$

$$v = 15.23 \text{ m/s}$$

Q9a) $\text{avg velocity} = \frac{\Delta \text{displacement}}{\Delta \text{time}}$

$$V_{\text{avg}} = \frac{25}{20}$$

$$V_{\text{avg}} = 1.25 \text{ m/s}$$

b) $\text{avg velocity} = \frac{\Delta \text{displacement}}{\Delta \text{time}}$

$$V_{\text{avg}} = \frac{-25}{15}$$

$$V_{\text{avg}} = -1.67 \text{ m/s}$$

c) $\text{avg velocity} = \frac{\Delta \text{displacement}}{\Delta \text{time}}$

$$V_{\text{avg}} = \frac{25 + (-25)}{20 + 15}$$

$$V_{\text{avg}} = \frac{0}{35}$$

$$V_{\text{avg}} = 0 \text{ m/s}$$

d) $\text{avg speed} = \frac{\text{distance}}{\text{time}}$

$$\text{avg speed} = \frac{25 + 25}{20 + 15}$$

$$\text{avg speed} = \frac{50}{35}$$

$$\text{avg speed} = 1.43 \text{ m/s}$$

Q10) Case 1

$$x = -3t^2 + 4t - 2$$

$$v_x = \frac{dx}{dt} = -6t + 4$$

$$a_x = \frac{dv_x}{dt} = -6$$

constant

$$y = 6t^2 - 4t$$

$$v_y = \frac{dy}{dt} = 12t - 4$$

$$a_y = \frac{dv_y}{dt} = 12$$

constant

∴ acceleration is constant, as both components are constant

Case 2

$$x = -3t^3 - 4t$$

$$v_x = \frac{dx}{dt} = -9t^2 - 4$$

$$a_x = \frac{dv_x}{dt} = -18t$$

not constant

$$y = -5t^2 + 6$$

$$v_y = \frac{dy}{dt} = -10t$$

$$a_y = \frac{dv_y}{dt} = -10$$

constant

∴ acceleration is not constant as ^x~~both~~ component is not constant

Case 3

$$r(t) = 2t^2 i - (4t+3) j$$

$$v_x = \frac{dx}{dt} = 4t$$

$$a_x = \frac{dv_x}{dt} = 4$$

constant

$$v_y = \frac{dy}{dt} = -4$$

$$a_y = \frac{dv_y}{dt} = 0$$

constant

∴ acceleration is constant as both component are constant.

Case 4

$$r = (4t^3 - 2t)i + 3j$$

$$V_x = \frac{dx}{dt} = 12t^2 - 2$$

$$a_x = \frac{dv_x}{dt} = 24t$$

not constant

$$V_y = \frac{dy}{dt} = 0$$

$$a_y = \frac{dv_y}{dt} = 0 \text{ constant}$$

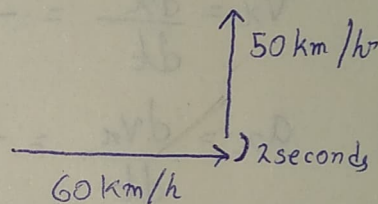
∴ acceleration is not constant as x component is not constant.

Q11) initial velocity = $60 \text{ km/hr } \hat{i}$
final velocity = $50 \text{ km/hr } \hat{j}$

in m/s

$$V_i = 60 \times \frac{1}{3.6} = 16.67 \text{ m/s } \hat{i}$$

$$V_f = 50 \times \frac{1}{3.6} = 13.89 \text{ m/s } \hat{j}$$



Change in velocity

$$\Delta v = V_f - V_i$$

$$\Delta v = (13.89)\hat{j} - (16.67)\hat{i}$$

$$\Delta v = -16.67\hat{i} + 13.89\hat{j}$$

$$|\Delta v| = \sqrt{(-16.67)^2 + (13.89)^2}$$

$$= 21.7 \text{ m/s}$$

for average acceleration $\Rightarrow$

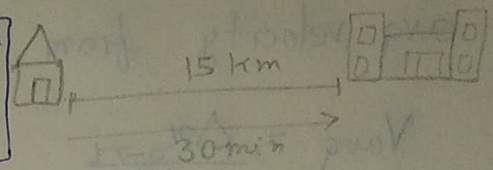
$$a_{\text{avg}} = \frac{|\Delta \vec{v}|}{\Delta t} = \frac{21.7}{2} = 10.85 \text{ m/s}^2$$

Q12a) $\text{avg speed} = \frac{\text{distance}}{\text{time}}$

$$= \frac{15}{0.5}$$

$\text{avg speed} = 30 \text{ km/hr}$

30 min
is
0.5 hr



$$\begin{array}{l} 60 \times \frac{1}{x} \\ 30 \end{array} \quad \left| \quad \begin{array}{l} 60 \times \frac{1}{x} \\ 20 \end{array} \right. \quad \begin{array}{l} x = \frac{1}{2} \text{ hr} \\ x = \frac{20}{60} = \frac{1}{3} \end{array}$$

b) $\text{avg speed} = \frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{15 + 15}{0.5 + \frac{1}{3}}$$

$H \rightarrow S = 30 \text{ min} = 0.5 \text{ hr}$
 $S \rightarrow H = 20 \text{ min} = \frac{1}{3} \text{ hr}$

$\text{avg speed} = 36 \text{ km/hr}$

c) $\text{avg velocity} = \frac{\text{Displacement}}{\text{Total time}}$

$$V_{\text{avg}} = \frac{15 - 15}{0.5 + \frac{1}{3}}$$

$V_{\text{avg}} = 0 \text{ km/hr}$

Q13a) Displacement from $t=0$ to $t=1$

$$x = -4t + 2t^2$$

$$\begin{array}{l} x = -4(0) + 2(0)^2 \\ x = 0 \end{array} \quad \left| \quad \begin{array}{l} x = -4(1) + 2(1)^2 \\ x = -2 \text{ m} \end{array} \right.$$

$\therefore \Delta x = -2 \text{ m}$

Displacement from $t=1$ to $t=3$

$$x = -4(3) + 2(3)^2$$

$$x = 6 \text{ m}$$

Total displacement = $6 - (-2)$
 $= 8 \text{ m}$

b) avg velocity from $t=0$ to $t=1$

$$V_{avg} = \frac{\Delta x_{0 \rightarrow 1}}{\Delta t}$$

$$V_{avg} = \frac{-2}{1}$$

$$V_{avg} = -2 \text{ m/s}$$

avg velocity from $t=1$ to $t=3$

$$V_{avg} = \frac{\Delta x_{1 \rightarrow 3}}{\Delta t}$$

$$V_{avg} = \frac{8}{2}$$

$$V_{avg} = 4 \text{ m/s}$$

c) instantaneous velocity

$$v(t) = \frac{dx}{dt} = -4 + 4t$$

at $t=2.5$

$$v(2.5) = -4 + 4(2.5)$$

$$v = 6 \text{ m/s}$$

Q14) a) $v = u + at$

$$0 = 20 + (-9.81)t$$

$$t = \frac{-20}{-9.81}$$

$$t = 2.038 \text{ s}$$

$$t \approx 2.04 \text{ s}$$

b) $v^2 = u^2 + 2as$

$$0 = 20^2 + 2(-9.81)s$$

$$s = \frac{-400}{-19.62}$$

$$s = 20.38 \text{ m}$$

$$\begin{aligned} \text{Total(s)} &= \text{Thrown(s)} + \text{Building s} \\ &= 20.4 + 50 \\ &= 70.4 \text{ m} \end{aligned}$$

c) $\text{Time} = 2(2.04)$
 $= 4.08 \text{ s}$

Q15)

$$V_y = 0$$

$$S_y = 100 \text{ m}$$

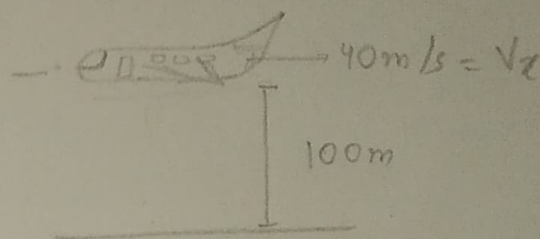
$$g = 9.81$$

$$S = ut + \frac{1}{2}at^2$$

$$100 = 0 + \frac{1}{2}(9.81)t^2$$

$$t = 4.515$$

$$t \approx 4.52 \text{ s}$$



Horizontal Distance

$$V_x = \frac{d}{t}$$

$$d = V_x t$$

$$= 40 \times 4.52$$

$$= 180.8 \text{ m}$$